

MANZANILLO, Mexico

WMO: 766540

Lat: 19.0444N

Lon: 104.3186W

Elev: 3

StdP: 101.29

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	18.5	19.4	10.8	8.1	22.3	12.7	9.1	22.7	6.6	25.2	6.0	25.4	1.5	0	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	33.6	26.5	33.1	26.5	32.5	26.5	28.4	31.8	28.0	31.6	27.6	31.2	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	30.9	26.9	22.6	30.7	26.5	22.1	30.4	91.5	32.0	89.4	31.7	87.6	31.3	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
6.1	5.1	4.3	DB	15.7	34.6	1.9	1.9	14.3	35.9	13.2	37.0	12.1	38.1	10.7	39.4	40
			WB	12.7	29.3	1.5	1.1	11.6	30.1	10.7	30.7	9.9	31.3	8.8	32.1	41

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	25.1	24.9	24.5	24.8	26.7	28.7	29.0	28.9	28.5	28.4	27.4	25.9
	DBStd	2.16	1.43	1.42	1.62	1.48	1.67	1.26	1.16	1.06	1.12	1.08	1.19	1.22
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6166	468	416	450	443	518	562	587	585	554	569	522	492
	CDD18.3	3124	210	183	192	193	259	312	329	327	304	311	272	233
	CDH23.3	31406	1481	1271	1303	1384	2527	3776	4093	3962	3530	3572	2709	1798
	CDH26.7	10880	356	276	279	288	752	1527	1747	1634	1339	1325	883	474
Wind	WSAvg	2.0	1.9	1.9	2.0	2.1	2.1	2.1	2.0	2.1	2.1	1.8	1.8	1.8
Precipitation	PrecAvg	990	10	13	0	0	1	110	172	212	271	130	21	15
	PrecMax	1529	119	199	9	6	19	333	590	710	767	586	280	94
	PrecMin	627	0	0	0	0	0	2	15	13	32	0	0	0
	PrecStd	271	24	37	2	1	4	93	135	144	159	119	52	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.0	31.2	31.5	32.7	34.2	34.2	34.4	33.6	33.2	32.6	31.4
		MCWB	22.6	23.9	22.9	23.3	25.6	25.7	26.9	26.0	26.6	26.9	25.6	24.4
	2%	DB	30.3	30.0	30.1	30.2	31.7	33.2	33.5	33.5	32.7	32.6	31.8	30.6
		MCWB	23.1	24.0	22.9	23.7	25.8	26.0	26.9	26.6	26.7	26.9	25.7	24.4
	5%	DB	29.5	29.2	29.2	29.3	30.8	32.4	32.8	32.8	32.1	31.9	31.1	30.0
		MCWB	23.3	23.7	23.1	23.8	25.6	26.1	26.7	26.7	26.7	26.8	25.4	24.1
	10%	DB	28.7	28.5	28.3	28.4	30.0	31.6	32.1	32.0	31.5	31.2	30.4	29.3
		MCWB	23.3	23.4	22.9	23.4	25.2	26.0	26.5	26.6	26.6	26.6	25.3	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	25.7	25.6	26.4	28.2	29.0	28.7	28.6	28.6	28.7	27.7	26.4
		MCDB	29.1	29.2	29.4	29.7	30.9	31.6	32.4	32.8	31.7	31.7	31.3	29.8
	2%	WB	25.0	25.1	24.7	25.4	27.1	28.0	28.1	27.9	27.9	28.1	27.1	25.8
		MCDB	28.6	28.9	28.7	29.0	30.2	31.1	32.0	32.1	31.4	31.3	30.5	29.3
	5%	WB	24.5	24.5	24.1	24.7	26.2	27.3	27.6	27.5	27.5	27.6	26.6	25.2
		MCDB	28.2	28.3	28.1	28.3	29.7	30.7	31.6	31.6	31.0	30.9	29.9	28.8
	10%	WB	23.9	23.9	23.5	24.0	25.6	26.7	27.1	27.1	27.1	27.1	26.1	24.7
		MCDB	27.7	27.6	27.4	27.6	29.2	30.4	31.1	31.2	30.6	30.4	29.4	28.2
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.4	7.6	7.4	6.2	5.5	6.0	6.2	5.7	5.8	6.5	7.1
		MCDBR	7.9	7.8	8.1	8.1	6.5	6.5	6.8	7.1	6.6	6.4	6.6	7.5
		MCWBR	5.3	5.1	5.1	5.1	3.7	3.1	3.5	3.8	3.3	3.3	3.5	4.7
	5% WB	MCDBR	7.3	7.3	7.5	7.5	6.2	5.5	6.3	6.6	6.1	5.9	6.0	6.8
MCWBR		5.4	5.3	5.2	5.2	4.2	3.2	3.7	3.9	3.4	3.5	3.7	4.5	
Clear-Sky Solar Irradiance	taub	0.386	0.386	0.376	0.425	0.530	0.511	0.477	0.467	0.464	0.423	0.386	0.388	
	taud	2.450	2.424	2.423	2.277	2.056	2.155	2.266	2.292	2.300	2.424	2.505	2.472	
	Ebn at Noon	893	913	932	883	785	793	821	835	838	865	887	879	
	Edh at Noon	106	115	118	138	171	153	137	134	132	113	100	101	
All-Sky Solar Radiation	RadAvg	4.86	5.80	6.90	7.43	7.10	6.04	5.77	5.66	5.12	5.16	5.01	4.65	
	RadStd	0.34	0.49	0.29	0.23	0.38	0.34	0.39	0.27	0.35	0.34	0.39	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)		+0.54	+1.41	-1.83	N/A	N/A	N/A	N/A	-107	+228	+130

Nomenclature: See separate page